

3. (a) The table gives the composition of six particles, **A-F**, which are either atoms or ions.

Particle	Number of protons	Number of neutrons	Number of electrons
A	14	14	14
B	19	20	18
C	15	16	18
D	16	16	16
E	11	12	11
F	12	12	10

- (i) Which particles are atoms? Explain your choice.

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- (ii) Which particles are positive ions? Give the charges on the particles you have chosen.

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- (b) Carbon has two isotopes – carbon-12 and carbon-14.

Using these examples, explain what is meant by the term *isotope*.

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